IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

CONTROL SYSTEMS

Wednesday, 1 May 2:30 pm

Time allowed: 2 hours

There are THREE questions on this paper.

Answer THREE questions.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : S.A. Evangelou

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CONTROL SYSTEMS – 2023/2024

Information for candidates:

Remark for the Students: the information provided here is much more than needed to answer the Exam Paper Questions; that is, the presence of a specific formula does not necessarily mean that this formula is required to answer the questions.

- **Bode Plots to answer Question 3**

Remark for the Students: In case drawing Bode plots is deemed helpful or necessary to answer Question 3), an additional answer book is provided that contains a few clean semi-logarithmic charts that should be used and should be attached to the main Answer Book.

- **Miscellaneous**

– *Settling-time* at 1% for asymptotically stable first-order systems: $t_{s,0.01} \simeq 4.6 \cdot \tau$ where τ denotes the time-constant (that is, the inverse of the pole with a sign-change).

– *Settling-time* at 1% for asymptotically stable second-order systems with complex-conjugate poles: $t_{s,0.01} \simeq \dfrac{4.6}{\xi\,\omega_n}$ where ω_n denotes the natural angular frequency and ξ the damping-ratio.

Information for candidates (continued):

Main Laplace Transforms and General Properties

$f(t)$	$F(s)$
$1(t)$	$\frac{1}{s}$
$t \cdot 1(t)$	$\frac{1}{s^2}$
$t^2 \cdot 1(t)$	$\frac{2}{s^3}$
$e^{\alpha t} \cdot 1(t)$	$\frac{1}{s-\alpha}$
$t \cdot e^{\alpha t} \cdot 1(t)$	$\frac{1}{(s-\alpha)^2}$
$\sin(\omega t) \cdot 1(t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t) \cdot 1(t)$	$\frac{s}{s^2+\omega^2}$
$t \cdot \sin(\omega t) \cdot 1(t)$	$\frac{2\omega s}{(s^2+\omega^2)^2}$
$t \cdot \cos(\omega t) \cdot 1(t)$	$\frac{s^2-\omega^2}{(s^2+\omega^2)^2}$
$e^{\sigma t} \cdot \sin(\omega t) \cdot 1(t)$	$\frac{\omega}{(s-\sigma)^2+\omega^2}$
$e^{\sigma t} \cdot \cos(\omega t) \cdot 1(t)$	$\frac{s-\sigma}{(s-\sigma)^2+\omega^2}$
$t \cdot e^{\sigma t} \cdot \sin(\omega t) \cdot 1(t)$	$\frac{2\omega(s-\sigma)}{\left[(s-\sigma)^2+\omega^2\right]^2}$
$t \cdot e^{\sigma t} \cdot \cos(\omega t) \cdot 1(t)$	$\frac{(s-\sigma)^2-\omega^2}{\left[(s-\sigma)^2+\omega^2\right]^2}$

Table 1 Time-functions and corresponding Laplace transforms

$\mathscr{L}[f(t)+g(t)]$	$F(s)+G(s)$
$\mathscr{L}\left[\int_0^t f(\tau)d\tau\right]$	$\frac{1}{s}F(s)$
$\mathscr{L}\left[\dot{f}(t)\right] = \mathscr{L}\left[\frac{d}{dt}f(t)\right]$	$sF(s)-f(0)$
$\mathscr{L}\left[\ddot{f}(t)\right] = \mathscr{L}\left[\frac{d^2}{dt^2}f(t)\right]$	$s^2F(s)-sf(0)-\dot{f}(0)$
$\mathscr{L}\left[\frac{d^3}{dt^3}f(t)\right]$	$s^3F(s)-s^2f(0) -s\dot{f}(0)-\ddot{f}(0)$
$\mathscr{L}\left[e^{kt}f(t)\right]$	$F(s-k)$
$\mathscr{L}[f(t-t_0)]$	$e^{-st_0}F(s)$
$\mathscr{L}[f(at)]\,, a \in \Re, a>0$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
$\mathscr{L}[t^n f(t)]$	$(-1)^n\frac{d^n}{ds^n}F(s)$
$\mathscr{L}[f(t)*g(t)] =$	
$= \mathscr{L}\left[\int_0^t f(\tau)g(t-\tau)d\tau\right]$	$F(s)G(s)$
If $\deg[N(s)] < \deg[D(s)]$ with $F(s)=N(s)/D(s)$	$f(0)=\lim_{s\to\infty}F(s)$
If all poles of $F(s)$ have negative real part except (possibly) one pole in $s=0$	$\lim_{t\to\infty}f(t)=\lim_{s\to 0}sF(s)$

Table 2 Main properties of Laplace tranforms

1. Consider the block scheme shown in Figure 1.1:

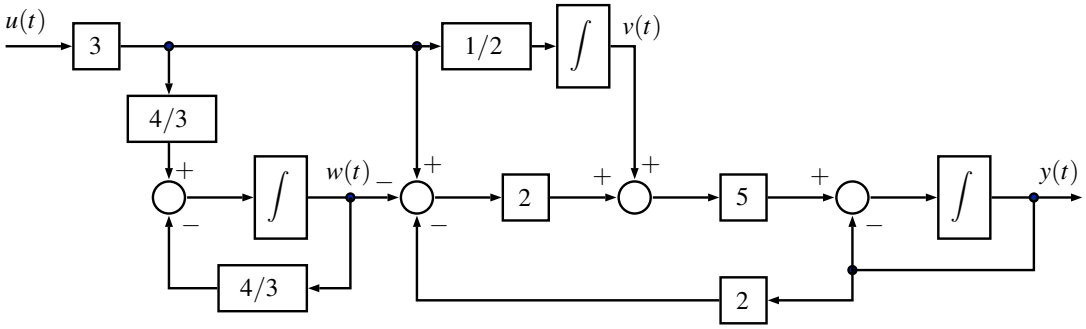


Figure 1.1 Block scheme for Question 1.

where $u(t) \in \mathbb{R}$ denotes the input function, and $y(t) \in \mathbb{R}$ denotes the output function.

a) Determine a state-space description of the dynamic system shown in Figure 1.1 of the form:

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases}$$

where $x \in \mathbb{R}^n$ is a state vector to be suitably defined, $\dot{x}$ denotes its time-derivative, u denotes the input, y denotes the output, and A, B, C, D are suitable matrices to be determined.

[15 Marks]

b) Using the matrix A obtained in your answer to Question 1-a), determine the matrix exponential e^{At}.

[15 Marks]

c) Owing to your answer to Question 1-b), setting $u(t) = 0, \forall t \in \mathbb{R}$, and setting the initial conditions of the integrator blocks shown in Figure 1.1 as

$$v(0) = 1; \quad w(0) = 0; \quad y(0) = -1,$$

determine the analytic expressions of the state vector function $x(t) \in \mathbb{R}^n, \forall t \in \mathbb{R}$ and of the output function $y(t), \forall t \in \mathbb{R}$.

[10 Marks]

2. Consider the block scheme shown in Figure 2.1:

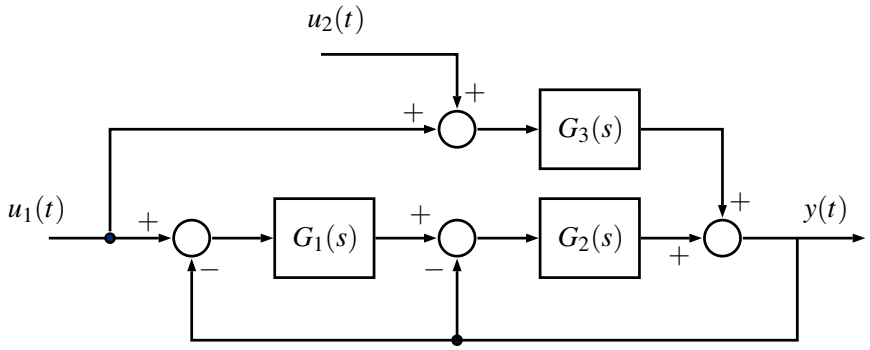


Figure 2.1 Block scheme for Question 2.

where, with some abuse of notation, we represent the inputs and outputs of each block in the time-domain instead of in the Laplace domain. The transfer functions $G_1(s)$, $G_2(s)$, and $G_3(s)$ are given by:

$$G_1(s) = \frac{5}{s+5}; \quad G_2(s) = 12; \quad G_3(s) = \frac{2s}{1-5s}$$

and it is assumed that $G_1(s)$, $G_2(s)$, and $G_3(s)$ have no hidden common factors in their numerators and denominators.

a) Determine the transfer function $G_{u_1,y}(s)$ from the input $u_1(t)$ to the output $y(t)$.

[10 Marks]

b) Determine the transfer function $G_{u_2,y}(s)$ from the input $u_2(t)$ to the output $y(t)$.

[6 Marks]

c) Establish the stability properties of the system described by the block scheme shown in Figure 2.1 and ascertain whether there is a relationship between these properties and the stability properties of the individual blocks $G_1(s)$, $G_2(s)$, and $G_3(s)$.

[4 Marks]

3. Consider the block scheme shown in Figure 3.1:

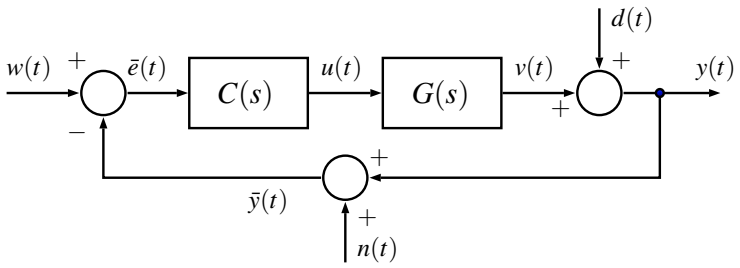


Figure 3.1 Block scheme for Question 3.

where, with some abuse of notation, we represent the inputs and outputs of each block in the time-domain instead of in the Laplace domain. Moreover

$$G(s) = 12 \cdot \frac{1 + 0.1\,s}{(1 + 20\,s)(1 + 3.3\,s)}$$

a) Setting $n(t) = 0, \forall t$, design a controller $C(s)$ such that the following Requirements i)–iv) are simultaneously satisfied:

i) The closed-loop steady-state static error $\bar{e}_\infty$ satisfies:

$$w(t) = A \cdot \mathbf{1}(t)\,,\ \forall A \in \mathbb{R};\ d(t) = B \cdot \mathbf{1}(t)\,,\ \forall B \in \mathbb{R} \quad \Longrightarrow \quad \bar{e}_\infty = 0$$

where $\mathbf{1}(t)$ denotes the unit step function.

ii) The crossover frequency ω_c satisfies $5\,\text{rad/sec} \leq \omega_c \leq 15\,\text{rad/sec}$

iii) The settling time at 1% satisfies $t_{s,0.01} \leq 2\,\text{sec}$

iv) The time-behaviour of the output $y(t)$ does not show any overshoot for $w(t) = A \cdot \mathbf{1}(t)\,,\ \forall A \in \mathbb{R};\ d(t) = 0\,,\ \forall t \in \mathbb{R}$

where $\mathbf{1}(t)$ denotes the unit step function. If needed or deemed useful, use the semi-logarithmic chart provided in the Additional Answer book.

[20 Marks]

b) Use the controller $C(s)$ that you have obtained in answering Question 3-a) and set $w(t) = 3 \cdot \mathbf{1}(t)$, $d(t) = 2 \cdot \mathbf{1}(t-5)$, and $n(t) = 0.5 \cdot \mathbf{1}(t-1)$, where $\mathbf{1}(t)$ denotes the unit step function. Determine analytically the output function $y(t)\,,\ \forall t$ and sketch its plot. Moreover, inspect the qualitative time-behaviour of the plot of $y(t)\,,\ \forall t$ and verify that it matches Requirement iv) in Question 3-a). Discuss your findings related to the effects of the disturbances $d(t)$ and $n(t)$.

[8 Marks]

c) Use the controller $C(s)$ that you have obtained in answering Question 3-a) and set $w(t) = 0, \forall t$, $d(t) = 5 \cdot \mathbf{1}(t)$, and $n(t) = 2 \cdot \sin(\overline{\omega} t) \cdot \mathbf{1}(t)$, where $\mathbf{1}(t)$ denotes the unit step function.

Determine *approximately* the high-frequency range $[\omega^-, +\infty)$ such that

$$|y_\infty(t)| < 10^{-1}\,, \quad \forall \overline{\omega} \in [\omega^-, +\infty)$$

where $y_\infty(t)$ denotes the regime (steady-state) output function.

If needed or deemed useful, use the semi-logarithmic chart provided in the Additional Answer book.

[12 Marks]